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LLM inference in C/C++

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llama.app

Contributing

Security policy

0 stars 0 forks 0 watching 1 branch 1 tag Activity

Public repository · Forked from ggml-org/llama.cpp

Repository navigation bar with filters, search, and actions like 'Go to file', 'Add file', 'Code', and 'Sync fork'.

ramicza cli : persist reasoning_content in chat history (ggml-org#26362) c629da5 · 33 minutes ago

.devops	devops : add llama in all docker images (gg...	2 months ago
.gemini	contributing: tighten AI usage policy (ggml-o...	8 months ago
.github	vulkan: update vulkan sdk to 1.4.357.0 (ggm...	yesterday
.pi/gg	pi : remove docs from system prompt (ggml...	2 months ago
app	app : allow --version, --licenses & --help (ggm...	2 months ago
benches	benches : add Nemotron 3 Nano on DGX Sp...	5 months ago
ci	common : fix env names to all have LLAMA_...	3 months ago
cmake	cmake : do not check for bin install dir (ggm...	3 months ago
common	chat : enable tool call in thinking for DS4 (gg...	11 hours ago
conversion	mtmd: add minicpmv46 downsample (ggml-...	5 hours ago
docs	SYCL: add oneMKL GEMM flash attention fo...	yesterday
examples	args: refactor mlock/mmap/directio into loa...	last week
ggml	vulkan: add POOL_1D op (ggml-org#25431)	yesterday
gguf-py	mtmd: add n_embd_head (ggml-org#26342)	yesterday
grammars	docs : fix typos in CUDA-FEDORA.md and gr...	2 months ago
include	llama : load MTP tensors only if they are real...	yesterday
licenses	refactor : remove libcurl, use OpenSSL when...	7 months ago
media	media : add transparent icon svg and png [n...	11 months ago
models	common/chat: add specialized minimax m3...	4 days ago
pocs	libs : rename libcommon -> libllama-commo...	4 months ago
requirements	model: add Mellum architecture (ggml-org#...	2 months ago

 scripts	sync : ggml	2 days ago
 skills	docs: add exception about weight folding (ggml-org#23871)	5 days ago
 src	Support rotated kv cache quant (ggml-org#23871)	yesterday
 tests	chat : enable tool call in thinking for DS4 (ggml-org#23871)	11 hours ago
 tools	cli : persist reasoning_content in chat histor...	33 minutes ago
 vendor	vendor: update BoringSSL to 0.20260728.0 (...)	3 days ago
 .clang-format	fix: apply clang-format to CUDA macros (ggml-org#23871)	11 months ago
 .clang-tidy	clang-tidy : disable warning about performa...	11 months ago
 .dockerignore	docker : prebuild web UI for s390x build [no ...]	2 months ago
 .ecrc	common : Update stb_image.h to latest vers...	2 years ago
 .editorconfig	ui: Restructure repo to use tools/ui folder a...	3 months ago
 .flake8	llama : move end-user examples to tools dir...	last year
 .gitignore	ui: PWA support (ggml-org#23871)	2 months ago
 .gitmodules	ggml : remove kompute backend (ggml-org#23871)	last year
 .pre-commit-config.yaml	convert.py : add python logging instead of p...	2 years ago
 AGENTS.md	skill: create add-new-model and code-revie...	last week
 AUTHORS	authors : update (ggml-org#19263)	6 months ago
 CLAUDE.md	contributing: tighten AI usage policy (ggml-org#23871)	8 months ago
 CMakeLists.txt	common: add subprocess.h wrapper, disabled ...	last week
 CMakePresets.json	cmake : Add CMake presets for Linux and G...	last year
 CODEOWNERS	HIP: remove rocWMMA FlashAttention (ggml-org#23871)	last week
 CONTRIBUTING.md	contrib : add guideline about the "merge rea...	4 days ago
 LICENSE	docs : Minor cleanups (ggml-org#19252)	6 months ago
 Makefile	make : remove make in favor of CMake (ggml-org#23871)	last year
 README.md	readme : refresh (ggml-org#26280)	2 days ago
 SECURITY.md	binaries : Improve rpc-server and export-gra...	2 months ago
 build-xcframework.sh	xcframework : disable mtmd video on i/tv/vi...	2 months ago
 convert_hf_to_gguf.py	convert_hf_to_gguf: support split MTP expo...	3 weeks ago
 convert_hf_to_gguf_update.py	Add support for Laguna XS.2 & M.1 (ggml-org#23871)	2 weeks ago
 convert_llama_ggml_to_gguf.py	ci : switch from pyright to ty (ggml-org#20871)	5 months ago
 convert_lora_to_gguf.py	convert : fix lora base model arch retrieval (...)	2 months ago
 flake.nix	fix(nix): remove non-functional llama-cpp ca...	last year
 mypy.ini	convert : partially revert PR ggml-org#4818 (...)	2 years ago
 pyproject.toml	model: add Mellum architecture (ggml-org#23871)	2 months ago
 pyrightconfig.json	ci : switch from pyright to ty (ggml-org#20871)	5 months ago

requirements.txt

tool-call: fix Qwen 2.5 Coder support, add ...

last year

ty.toml

mtmd : DeepSeek-OCR image processing fix...

3 months ago

README

Contributing

License

Security



llama.cpp



LLM inference in C/C++

license MIT release b10218 Server passing Publish Docker image failing Update Winget Package passing

[manifesto](#) / [ggml](#) / [ops](#) / [maintainer PRs](#) / [dev branches](#) / [compile times](#) / [lib llama API](#) / [llama-server REST API](#)

Quick start

A few options to get llama.cpp installed on your machine:

- Visit <https://llama.app> and follow the instructions
- Run with Docker - see our [Docker documentation](#)
- Download pre-built binaries from the [releases page](#)
- Build from source by cloning this repository - check out [our build guide](#)

Once installed:

```
# Download and run a model directly from Hugging Face
llama cli -hf ggml-org/Qwen3.5-0.8B-GGUF

# Launch OpenAI-compatible API server
llama serve -hf ggml-org/Qwen3.5-0.8B-GGUF
```



```
llama cli -hf ggml-org/Qwen3.6-72B-GGUF-Q8_0
Loading model...

llama.cpp

build      : b10109-c0bc859fe
model      : ggml-org/Qwen3.6-72B-GGUF-Q8_0
type       : Q8_0
modalities : text, vision, video

Available commands:
  /api or /cli/c  stop or wait
  /api            regenerate the last response
  /clear         clear the chat history
  /read <file>   add a text file
  /gpt <system>  add text file using globbing pattern
  /image <file>  add an image file
  /video <file>  add a video file

> /image ./media/mtmul.png
Loaded media from './media/mtmul.png'
> /read ./CONTRIBUTING.md
Loaded text from './CONTRIBUTING.md'

> Explain the mt mul diagram.
Based on the diagram and the accompanying text in the 'CONTRIBUTING.md' file, here is an explanation of the matrix multiplication visualization.
See the Core Concept
The diagram explains a specific quirk in the 'llama.cpp' (specifically the 'ggml' library) implementation of matrix multiplication.
In standard linear algebra, if you want to calculate  $C = A \times B$ , you multiply matrix  $A$  by matrix  $B$ .
However, in ggml, the function 'ggml_mul_mat(ctx, A, B)' actually computes:

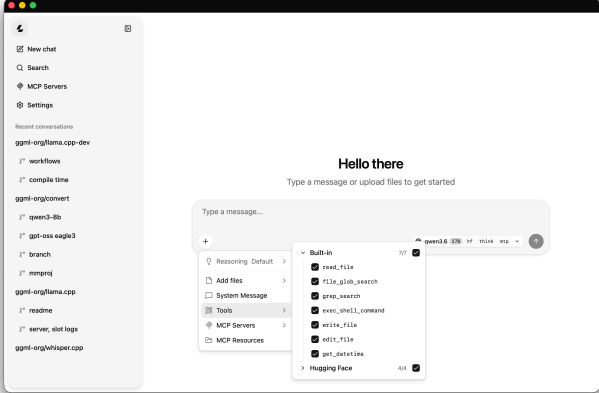
$$C^T = A \times B^T$$

which implies:

$$B^T C = (A \times B^T)^T = B \times A^T$$

```

VLM session with llama cli



Built-in web UI against llama serve

Description

The main goal of llama.cpp is to enable LLM (and VLM) inference with minimal setup and state-of-the-art performance on a wide range of hardware - locally and in the cloud.

- Plain C/C++ implementation without any dependencies
- Apple silicon is a first-class citizen - optimized via ARM NEON, Accelerate and Metal frameworks
- AVX, AVX2, AVX512 and AMX support for x86 architectures
- RVV, ZVFH, ZFH, ZICBOP and ZIHINTPAUSE support for RISC-V architectures
- 1.5-bit, 2-bit, 3-bit, 4-bit, 5-bit, 6-bit, and 8-bit integer quantization for faster inference and reduced memory use
- Custom CUDA kernels for running LLMs on NVIDIA GPUs (support for AMD GPUs via HIP and Moore Threads GPUs via MUSA)
- Vulkan and SYCL backend support
- CPU+GPU hybrid inference to partially accelerate models larger than the total VRAM capacity

The llama.cpp project is build on top of the [ggml](#) library.

Supported backends

Backend	Target devices
BLAS	All
BLIS	All
CANN	Ascend NPU
CUDA	Nvidia GPU
HIP	AMD GPU
Hexagon [In Progress]	Snapdragon
IBM zDNN	IBM Z & LinuxONE
MUSA	Moore Threads GPU
Metal	Apple Silicon
OpenCL	Adreno GPU
OpenVINO [In Progress]	Intel CPUs, GPUs, and NPUs
RPC	All
SYCL	Intel GPU
VirtGPU	VirtGPU APIR

Backend	Target devices
Vulkan	GPU
WebGPU	All
ZenDNN	AMD CPU

[Documentation](#)

[Tools](#)

- [cli](#)
- [completion](#)
- [server](#)
- [GBNF grammars](#)

[Development](#)

- [How to build](#)
- [Running on Docker](#)
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[Contributing](#)

- Contributors can open PRs

Releases

 1 tag

[Create a new release](#)

Packages 1

 llama.cpp

Contributors

No contributors

Languages

C++ 55.1% C 16.5% Python 7.1% Cuda 5.6% TypeScript 4% HTML 2.3% Other 9.4%